Elements of AIML Lab
Experiment 03 (After-submission Reference): WEKA for Clustering

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1 How to use this reference

This document is meant to be shared **after you submit** Experiment 03. Use it to cross-check that your work demonstrates:

- clustering as an **unsupervised** task,
- at least two clusterers compared on the **same** dataset,
- the effect of **scaling/normalization** on distance-based clustering,
- an interpretation of what each cluster might represent.

2 Quick recap: what clustering is (and is not)

Clustering groups similar instances **without labels**. Clusters are **not guaranteed** to match human categories; they are patterns discovered from the features.

2.1 Why scaling matters

Many clustering methods use distances (e.g., Euclidean distance). If one feature has a much larger range, it dominates the distance and can change the cluster structure.

3 Worked example in WEKA (end-to-end)

3.1 Dataset choice (example)

If you need a dataset, WEKA often ships sample datasets under `data/`. Numeric datasets work well for k-means. If your dataset includes a label column, you may keep it for *interpretation only* (clustering itself does not use it).

3.2 Step 1: load data and remove obvious IDs

1. WEKA → **Explorer** → **Preprocess** → **Open file...**
2. Inspect attributes and remove ID-like columns (if any).

3.3 Step 2: run k-means once without scaling

1. Go to **Cluster** tab.
2. **Choose** → `SimpleKMeans`.
3. Set **NumClusters** (choose a reasonable k, e.g., 3 for a first run).
4. Click **Start**.

Record:

- chosen k,
- cluster sizes,
- centroid summaries (if shown),
- any available within-cluster error measure reported by WEKA.

3.4 Step 3: apply scaling and rerun k-means

1. **Preprocess** tab → **Filter**.
2. Try `unsupervised/attribute/Standardize` (or `Normalize`).
3. Apply the filter.
4. Return to **Cluster** tab and run `SimpleKMeans` again (same k first).

Cross-check: cluster sizes/centroids often change after scaling if feature ranges differ.

3.5 Step 4: run a second clusterer (comparison)

Choose one more:

- `EM` (probabilistic),
- `HierarchicalClusterer` (hierarchical).

Run it on the same (preferably scaled) dataset and compare:

- cluster sizes,
- stability/interpretability,
- whether the grouping aligns with your domain intuition.

3.6 Step 5: visualize cluster assignments

If available, use:

- **Visualize cluster assignments** to view 2D projections.

Take at least one screenshot and write 2–3 lines describing what the visualization suggests.

4 How to write interpretations (examples)

Good interpretation connects clusters to real meaning:

- "Cluster 1 has high income and low spending score; likely *savers*. Cluster 2 has moderate income and high spending score; likely *active shoppers*." (example pattern)
- "After scaling, clusters became more balanced; before scaling, the largest-range feature dominated grouping."

5 Troubleshooting (common issues and fixes)

- **Meaningless clusters:** remove IDs, scale numeric features, and re-check k.
- **Choosing k randomly:** justify using a simple heuristic (try multiple k values and compare within-cluster error + interpretability).
- **Treating clusters as labels:** clusters are exploratory; report them as patterns, not truth.

6 Report structure (1–2 pages)

1. Dataset (features used, rows/columns, any preprocessing).
2. Clusterer 1 results (k-means): k, cluster sizes, centroid summary highlights.
3. Scaling comparison: what changed after `Standardize/Normalize`.
4. Clusterer 2 results (EM/hierarchical) and comparison.
5. Interpretation: what each cluster could represent.

Sources: Provost and Fawcett, *Data Science for Business.*;McKinney, *Python for Data Analysis.*;WEKA Documentation (Waikato Environment for Knowledge Analysis), accessed 2026-02-28.;Matplotlib and Seaborn documentation, accessed 2026-02-28.